

CMSE 201 Final Project, Section 001

Important Dates:

- April 15: Optional In-Class Period; Come to class to work on your project and ask questions of instructors
 - April 19: Submit recorded presentations in *.mp4 format to D2L to the "Semester Project Video Presentation" assignment by 1:00pm
 - April 20: Submit Project **Jupyter notebook** and pdf of your slides to D2L to the "Semester Notebook and Slides" Assignment by midnight
 - April 20-22: Watch the recordings from your two assigned groups on Mediaspace; Ask at least one question of each project
 - April 23: By this date, answer majority of the questions posed on your own video.
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Overall requirements

For your semester project, **you'll be expected to turn in a 4-5 minute recorded presentation, the slides used in the presentation, and a Jupyter notebook file.**

Your **Jupyter notebook** should contain the following:

- All of the code used to complete your project. The code should be documented so that the instructor can understand what you did and why you did it. This means including meaningful comments in the code itself! You should also use logical variable names to ease the reading of the code.
- Narrative text (using Markdown cells) that explains what you did, why you did it, and how you went about it. You should break the notebook up into the following meaningful sections so that the instructor can follow the logical flow of your work:
 - **Background and Motivation** (Provide some context for the problem and what question(s) you set out to answer.)
 - **Methodology** (How did you go about answering your question(s)? Most of your code will be contained in this section.)

- **Results** (What did you find when you carried out your methods? Some of your code related to presenting results/figures/data may be replicated from the methods section or may only be present in this section. All of the plots that you plan on using for your presentation should be present in this section)
- **Synthesis and Discussion** (What did you learn from your results? What obstacles did you run into? What would you do differently next time? What is the answer to your question(s) and why?)
- **References** Include a reference/bibliography section to cite all sources of data, code, and information relevant to your project

The code, text, and visuals should be threaded together as coherent report of your work!

In addition to the notebook, **you will be recording a 4-5 minute presentation** using a short set of slides to share the results of your project with your fellow classmates and your instructors. Your recording should include appropriate **visual aids**. The presentation included in your recording should address:

- The **question(s)** you set out to answer.
- The **model** that you applied to your chosen data or chosen topic.
- The **computational techniques** that you used to analyze your data or run your model.
- The **answer(s)** you arrived at.

Presentation slides

You can create your slides using the presentation software of your choosing (e.g. Google Slides, Powerpoint, Keynote). Once you've finished your slides, you should save a PDF copy that you can submit with your Jupyter notebook. You should aim for having a reasonable number of slides -- a good rule of thumb is ~1 min of time per slide. For a presentation of this length, you probably wouldn't want to have more than 6-7 slides, but you should make sure that you have a sufficient number of slides to support your presentation.

The slides should:

- Address the above points in a logical order.
- Include a title slide with your project title, name, and course number (with section number).
- Effectively communicate information through a combination of text and figures. Text and figures should be clearly legible in your recorded presentation.
- Only include text and figures that support the presentation (avoid including content "just because").
- Avoid large paragraphs of text, use bullet points instead.

If you would like to try to make presentation slides using your Jupyter Notebook visit this [site](#). Note that this is **not** a requirement, but a nice feature that some instructors have found useful and interesting.

Recording

Once you're finished your presentation slides, you should use Zoom to record your presentation. You should be able to "Share Screen" in Zoom to share your slides and talk through the presentation as if you were giving it live. **You should make sure that your recording is no longer than 5 minutes, but it should be at least 4 minutes.** You will turn in the "mp4" file that Zoom produces when you record the Zoom session.

Grading Rubric

The presentation/project will be graded as follows:

- One fifth of the grade (**20 points**) comes from the oral presentation. Is it clear and well-structured? Does the student effectively communicate the key ideas about their results?
 - *Excellent presentation*: student speaks clearly and in a logical manner, and makes their key points clear. (**25 points**)
 - *Good presentation*: student speaks clearly and logically and conveys key points, but presentation is somewhat lacking (**19 points**)
 - *Fair presentation*: student's oral presentation is substantially lacking (speaking too quietly to be heard or with little inflection, clarity/logical flow in speaking is sub-par, etc.) (**13 points**)
 - *Poor presentation*: student's oral presentation is completely lacking (cannot be heard, there is no logical progression to the presentation) (**7 points**)
- One fifth of the grade (**20 points**) comes from the student's slides. Do the slides complement their oral presentation, and conform to the guidelines?
 - *Excellent slides*: Slides conform to specifications in final presentation document, including number of slides, adequately addressing all points, legibility. Slides complement oral presentation. (**25 points**)
 - *Good slides*: Slides deviate from specifications in some minor way: too many or too few slides, not all points addressed, some slides hard to understand (poor graphics, too much or too little text, etc.) (**19 points**)
 - *Fair slides*: Slides deviate from specifications in some substantial way: too many or too few slides, half or fewer of points addressed, most slides hard to understand (poor

graphics, too much or too little text, etc.) (**13 points**)

- *Poor slides*: Slides do not conform to specifications in any meaningful way. (**7 points**)
- One half of the grade (**50 points**) comes from the content of the project as evidenced in the Jupyter notebook and the presentation. Does the student clearly and concisely address all off the points listed in the final project guidelines?
 - *Excellent content*: All points clearly and concisely addressed. (**50 points**)
 - *Good content*: Most of the points clearly and concisely addressed, or all points nominally addressed but some are unclear. (**38 points**)
 - *Fair content*: Roughly half of the points clearly and concisely addressed, or all points nominally addressed but half are unclear. (**26 points**)
 - *Poor content*: Few of the points clearly and concisely addressed, or most/all points nominally answered but most are unclear. (**14 points**)
- One tenth of the grade (**10 points**) comes from peer questions and answers.
 - *Excellent Q/A*: Asked thoughtful questions of assigned presentations and responded to all questions on your presentation (**10 points**)
 - *Fair Q/A*: Asked thoughtful questions on fraction of presentations and/or asked poor questions on all presentations (**5 points**)